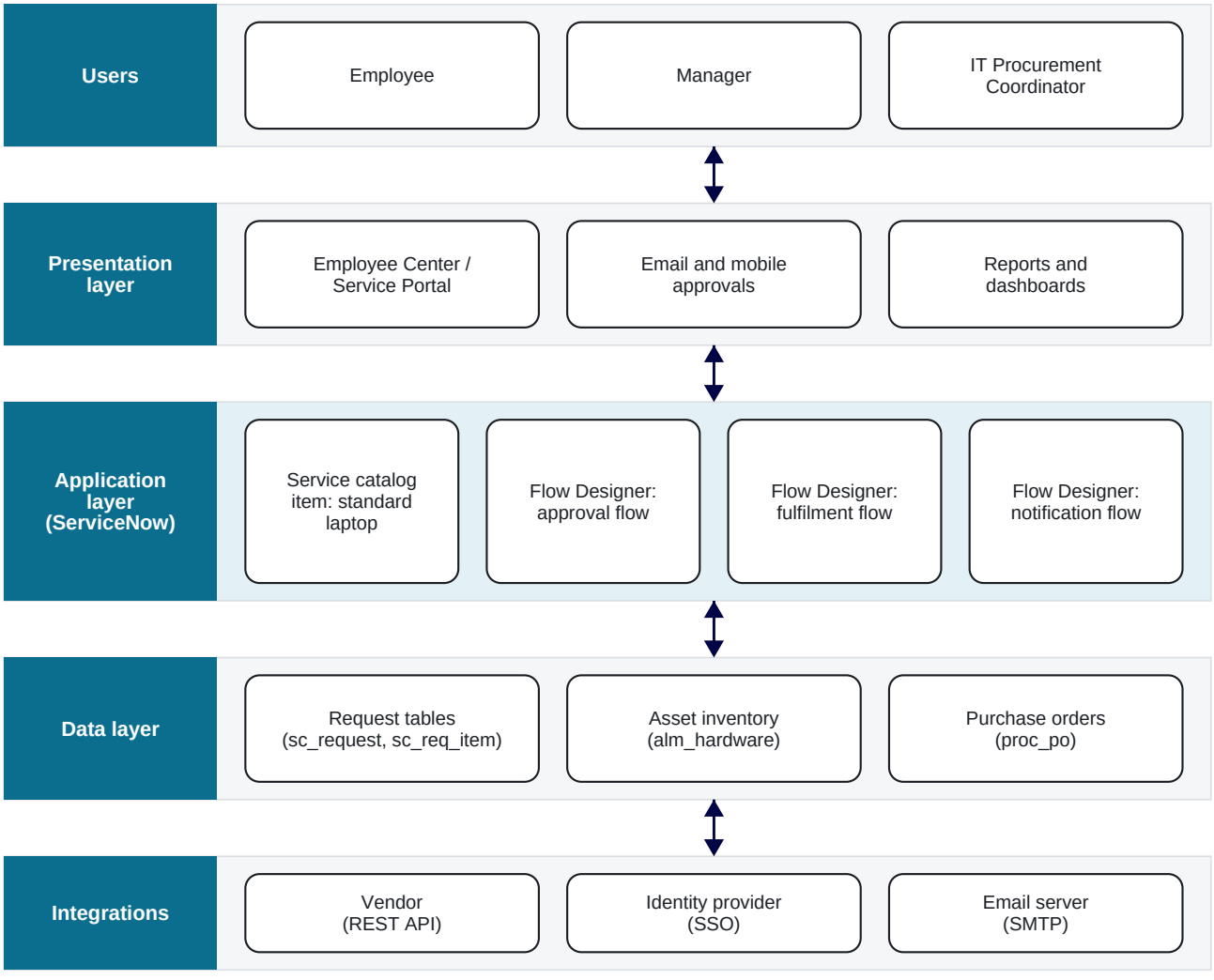
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 2025
Team ID	
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture shows the main layers of the solution and how they connect. Users reach the solution through the portal, email and mobile. Requests are handled by the ServiceNow Service Catalog and Flow Designer, which store and read data in platform tables and connect to outside systems such as the vendor.



Flow Designer runs the automation in the application layer: approvals, stock checks, purchase orders, asset records and notifications.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User interface	Employees request laptops and track status; managers and coordinators work on their tasks	ServiceNow Employee Center / Service Portal
2	Service catalog	Standard laptop catalog item with pre-filled variables and eligibility checks	ServiceNow Service Catalog, Catalog Item
3	Workflow automation	Approval, fulfilment and notification flows that run without manual steps	Flow Designer, Subflows, Actions
4	Approvals	Automatic approval for eligible roles and manager approval with escalation	Flow Designer approval action, Approval Engine
5	Database	Stores requests, assets and purchase orders	ServiceNow tables (sc_request, sc_req_item, alm_hardware, proc_po)
6	Notifications	Stage-by-stage updates to the requester and approvers	ServiceNow Notifications, Email, Mobile
7	External integration	Sends purchase orders to the assigned vendor	REST Message, IntegrationHub
8	Reporting	Turnaround time and savings dashboard	ServiceNow Reports, Performance Analytics
9	Cloud infrastructure	Hosts the application and database	ServiceNow cloud instance

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Low-code framework	Automation is built with a visual designer instead of custom code	Flow Designer, ServiceNow Studio
2	Security implementations	Role-based access, single sign-on and encrypted connections	ACLs and roles, SAML SSO, HTTPS
3	Scalable architecture	New models, roles and vendors are added through configuration	Subflows, catalog variables, decision tables
4	Availability	Hosted on a managed cloud platform	ServiceNow cloud infrastructure
5	Performance	Routine steps run automatically in the background, reducing manual delay	Flow Designer asynchronous execution

Screenshots

Screenshots showing the Service Catalog and Flow Designer layers of the architecture as configured in the ServiceNow instance.

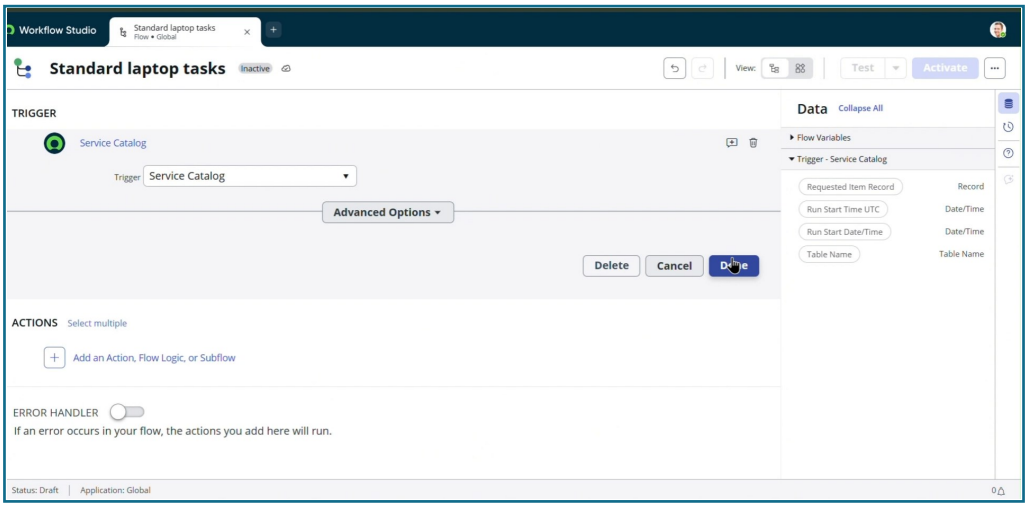


Figure 1: Flow Designer flow with the Service Catalog trigger

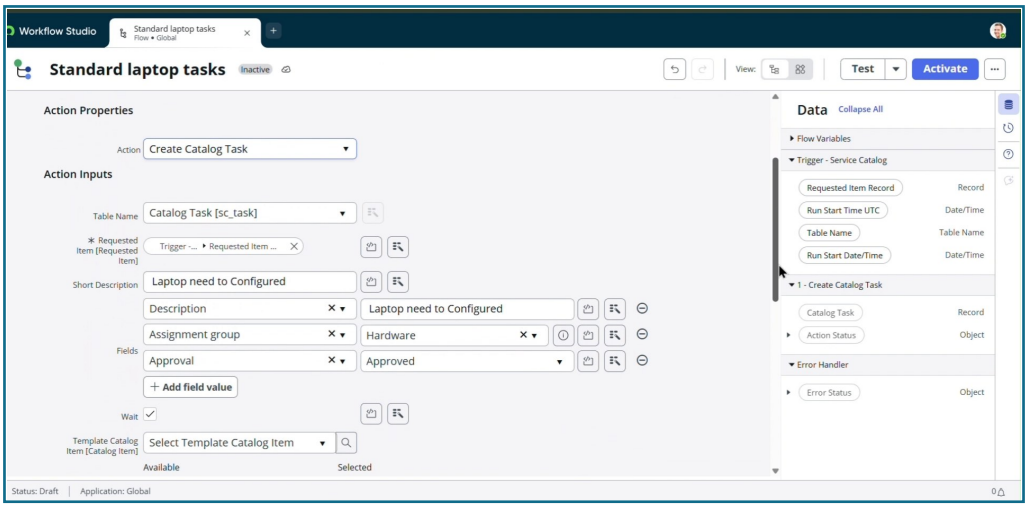


Figure 2: Create Catalog Task action configured in the flow

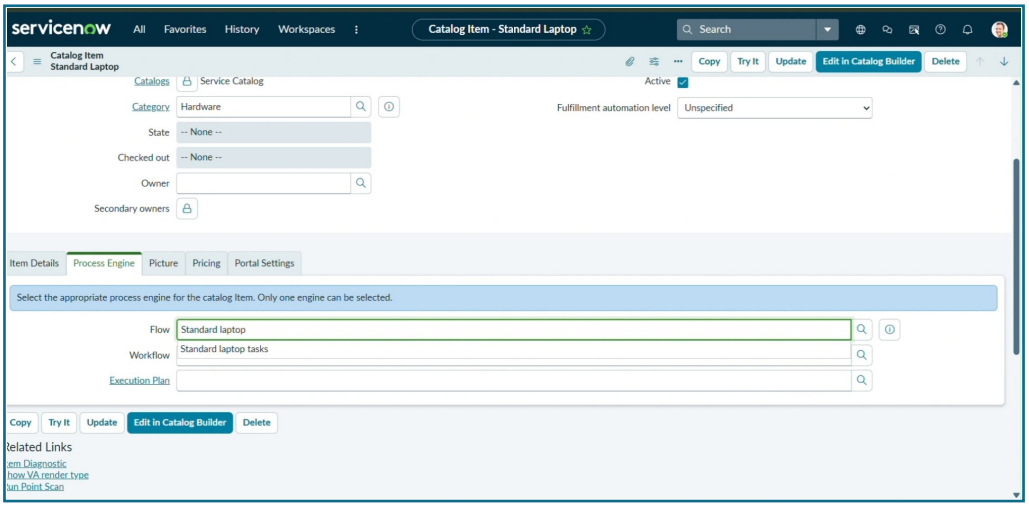


Figure 3: Standard Laptop catalog item linked to the "Standard laptop tasks" flow